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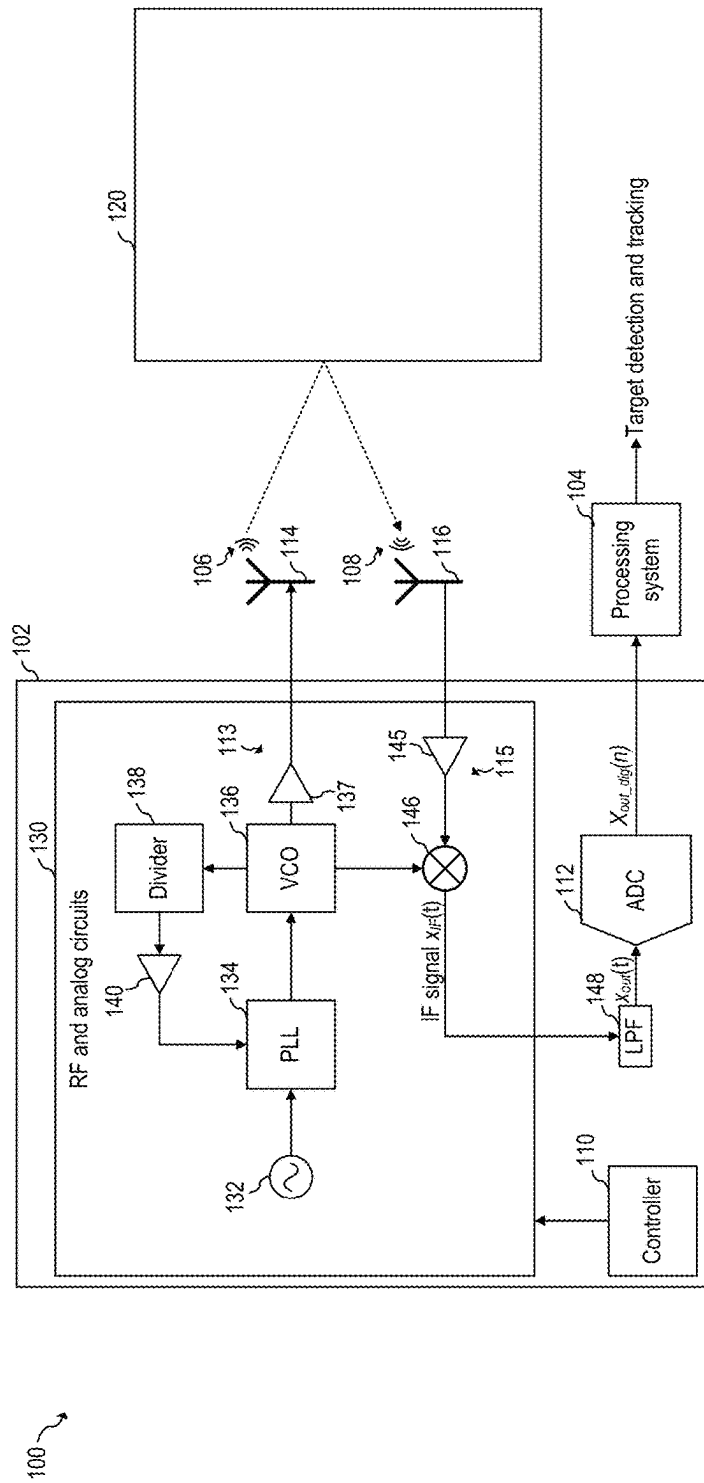
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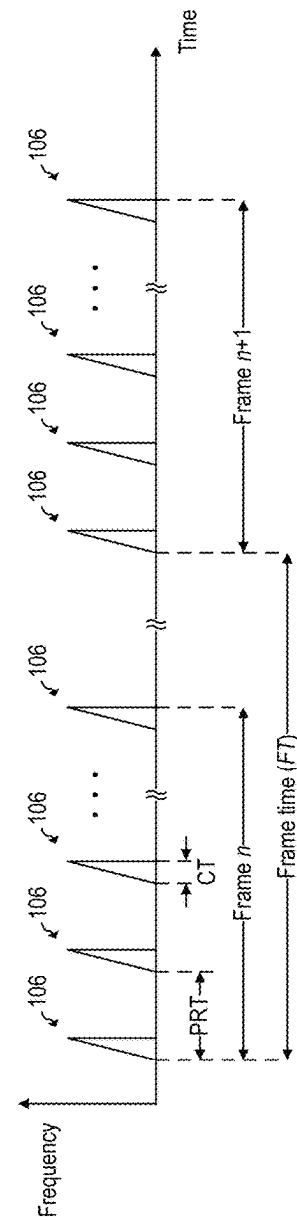


FIG. 2

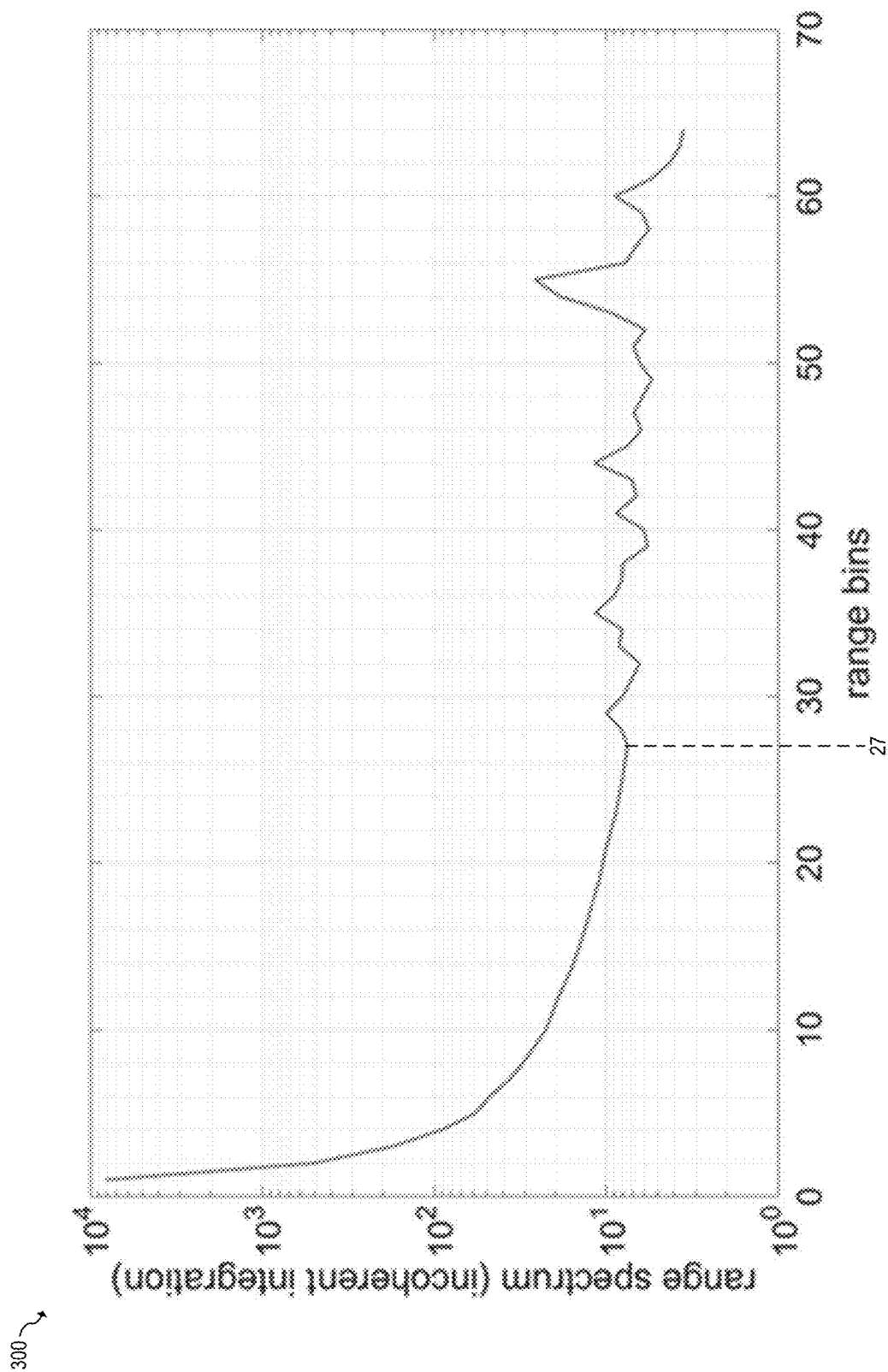


FIG. 3

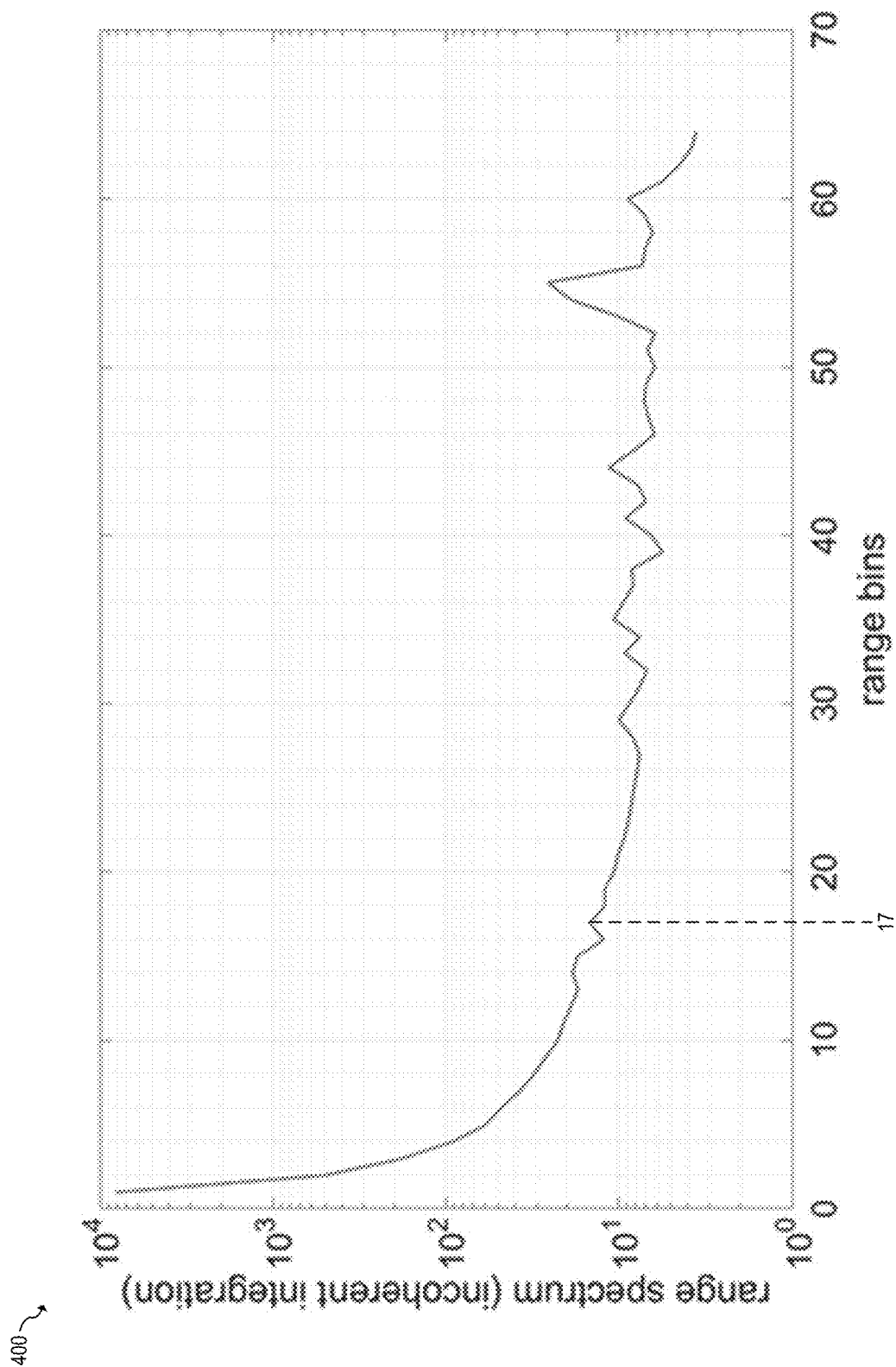


FIG. 4

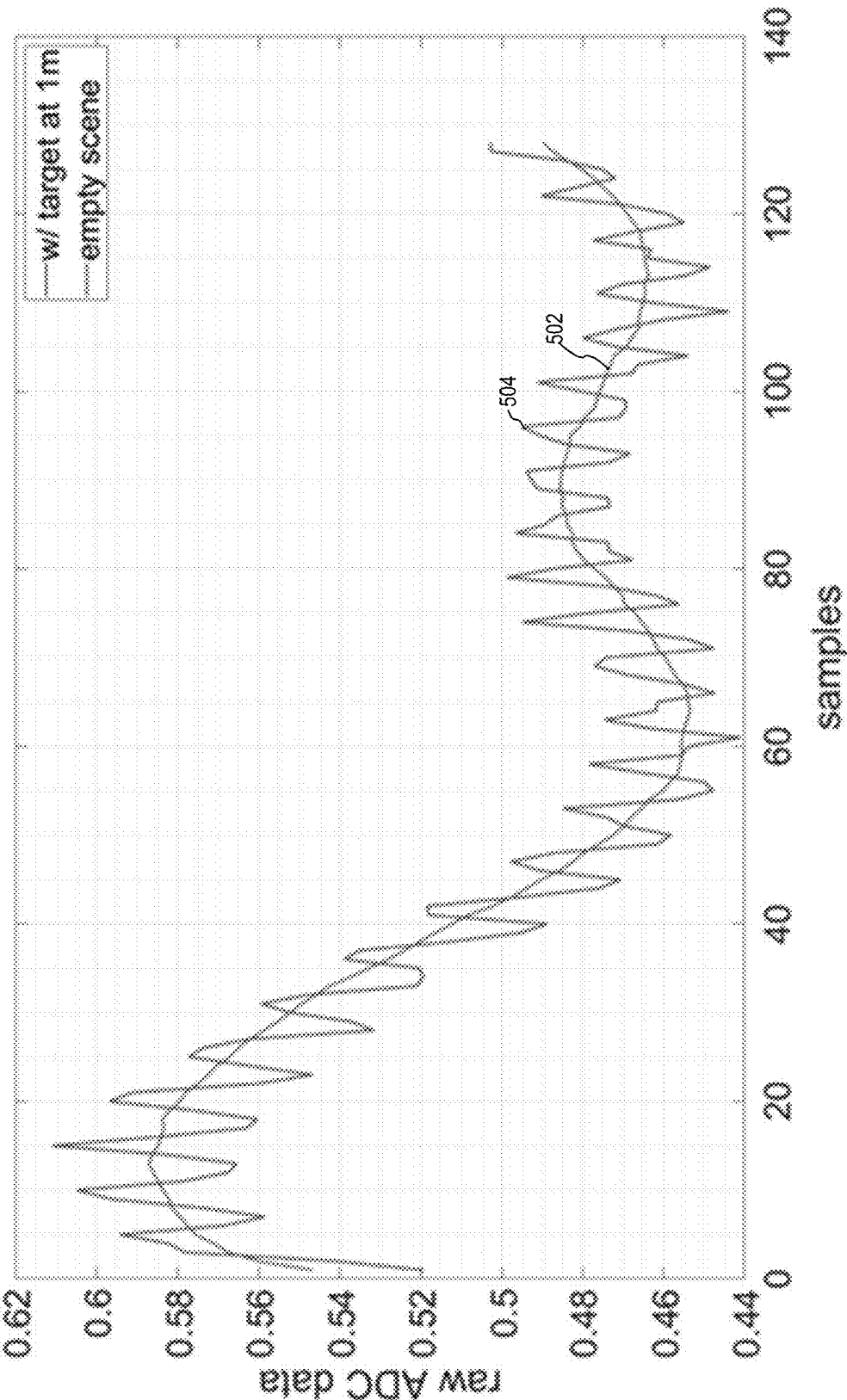


FIG. 5

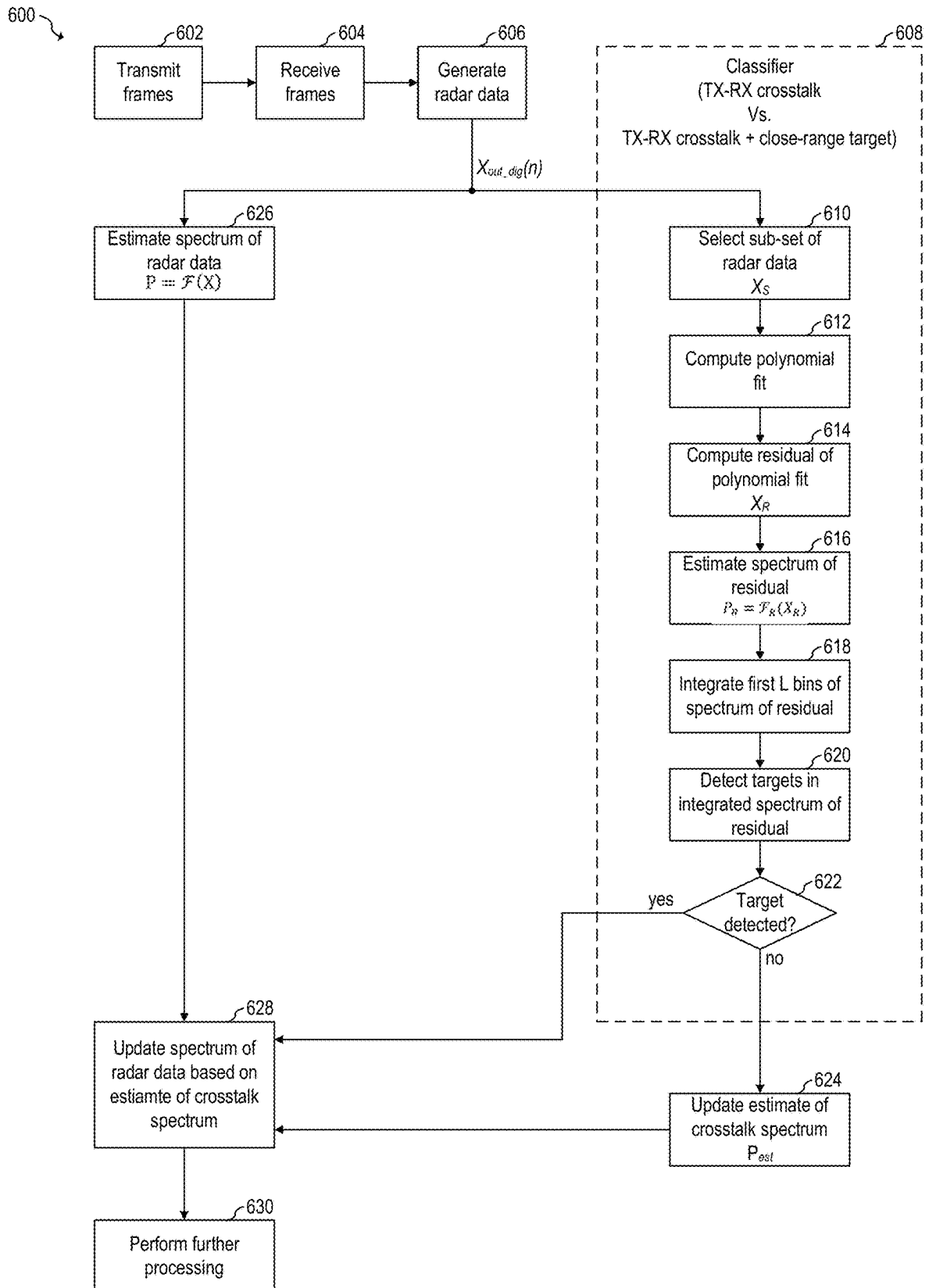


FIG. 6

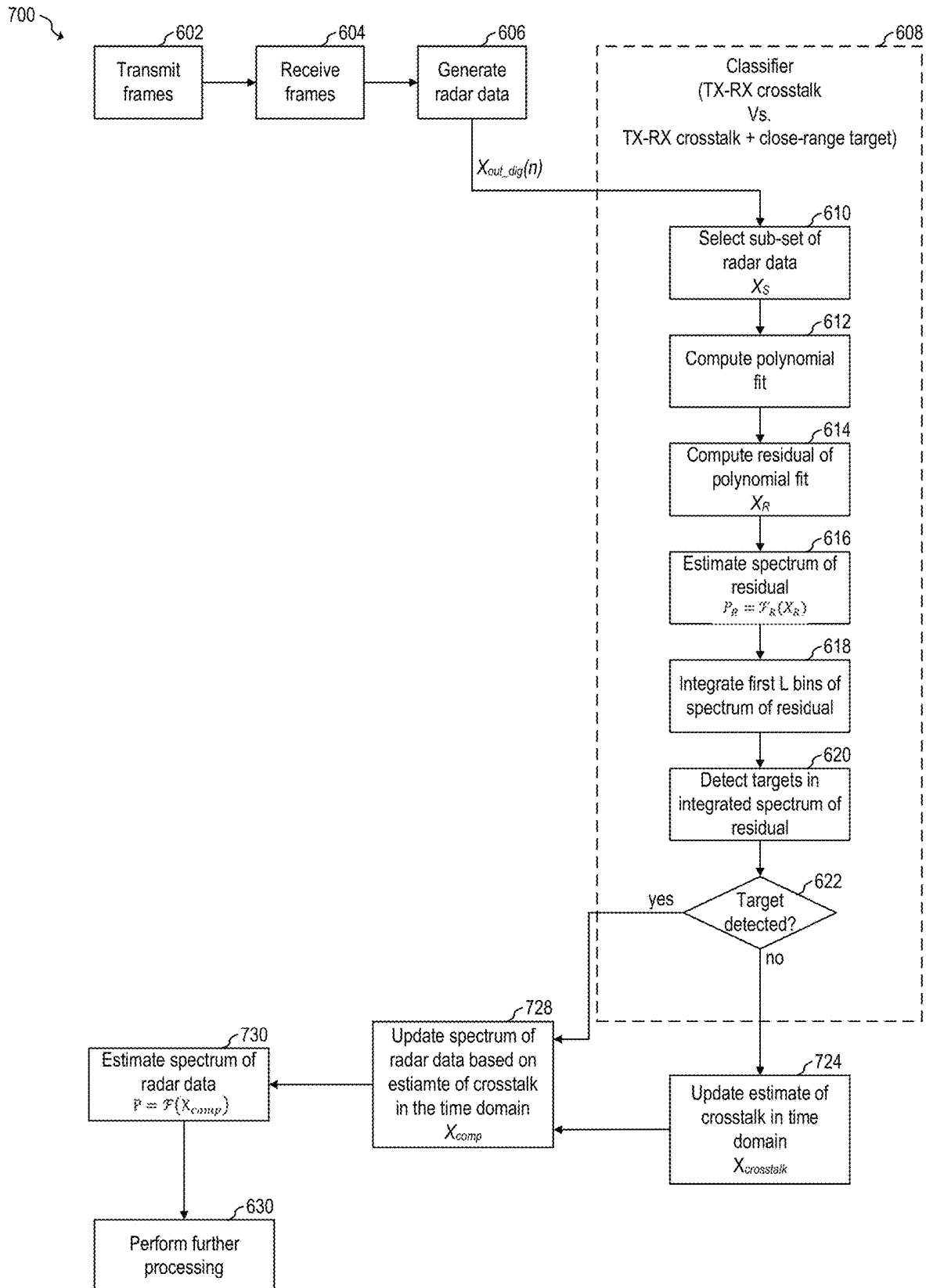


FIG. 7

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ADAPTIVE Tx-Rx CROSSTALK CANCELLATION FOR RADAR SYSTEMS

TECHNICAL FIELD

The present disclosure relates generally to an electronic system and method, and, in particular embodiments, to adaptive transmitter-receiver crosstalk cancellation for radar systems.

BACKGROUND

Applications in the millimeter-wave frequency regime have gained significant interest in the past few years due to the rapid advancement in low-cost semiconductor technologies, such as silicon germanium (SiGe) and fine geometry complementary metal-oxide semiconductor (CMOS) processes. Availability of high-speed bipolar and metal-oxide semiconductor (MOS) transistors has led to a growing demand for integrated circuits for millimeter-wave applications at e.g., 24 GHz, 60 GHz, 77 GHz, and 80 GHz and also beyond 100 GHz. Such applications include, for example, automotive radar systems and multi-gigabit communication systems.

In some radar systems, the distance between the radar and a target is determined by transmitting a frequency modulated signal, receiving a reflection of the frequency modulated signal (also referred to as the echo), and determining a distance based on a time delay and/or frequency difference between the transmission and reception of the frequency modulated signal. Accordingly, some radar systems include a transmit antenna to transmit the radio-frequency (RF) signal, and a receive antenna to receive the reflected RF signal, as well as the associated RF circuits used to generate the transmitted signal and to receive the RF signal. In some cases, multiple antennas may be used to implement directional beams using phased array techniques. A multiple-input and multiple-output (MIMO) configuration with multiple chipsets can be used to perform coherent and non-coherent signal processing as well.

Crosstalk from the transmit path of a radar to the receive path of the radar may occur, e.g., when a portion of the transmitted signal is received by the receive path via an internal path of the radar. Such crosstalk is commonly referred to as TX-RX crosstalk or TX-RX leakage. A standard approach to cope with this crosstalk is to use high-pass filters (HPFs). Another approach uses a moving target indication (MTI) in applications with moving targets.

SUMMARY

In accordance with an embodiment, a method includes: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; and after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

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In accordance with an embodiment, a radar system includes: a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of the receive path; and a processing system configured to: initialize a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path, select a set of data from the radar data, perform target detection on the set of data, and after performing the target detection on the set of data, when no target is detected in the set of data, update the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generate a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generate the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

In accordance with an embodiment, a method includes: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the a transmit path of a millimeter-wave radar sensor and a receive path of the millimeter-wave radar sensor; receiving radar data from the millimeter-wave radar sensor; selecting a sub-set of data from the radar data; determining a polynomial fit on the sub-set of data; determining a residual of the polynomial fit based on the polynomial fit and the sub-set of data; and determining a spectrum of the residual of the polynomial fit; incoherently integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than a maximum number of bins of the spectrum of the residual of the polynomial fit; performing target detection on the sub-set of data based on the integrated spectrum; after performing the target detection on the sub-set of data, when no target is detected in the sub-set of data, updating the crosstalk compensation factor based on the sub-set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the sub-set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; and performing target detection or target tracking based on the radar spectrum.

In accordance with an embodiment, a method includes: receiving radar data from a radar sensor; selecting a set of data from the radar data; determining a crosstalk approximation to a transmitter-receiver crosstalk between a transmit path of the radar sensor and a receive path of the radar sensor based on the set of data; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; determining a spectrum of the residual of the crosstalk approximation; and performing target detection on the set of data based on the spectrum of the residual of the crosstalk approximation.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

FIG. 1 shows a schematic diagram of a millimeter-wave radar system, according to an embodiment of the present invention;

FIG. 2 shows a sequence of chirps transmitted by the radar system of FIG. 1, according to an embodiment of the present invention;

FIGS. 3 and 4 show exemplary range spectra associated with the millimeter-wave radar sensor of FIG. 1, according to embodiments of the present invention;

FIG. 5 shows waveforms of radar data, according to an embodiment of the present invention; and

FIGS. 6 and 7 show flow charts of embodiment methods for dynamic crosstalk cancellation, according to an embodiment of the present invention.

Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

The making and using of the embodiments disclosed are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

The description below illustrates the various specific details to provide an in-depth understanding of several example embodiments according to the description. The embodiments may be obtained without one or more of the specific details, or with other methods, components, materials and the like. In other cases, known structures, materials or operations are not shown or described in detail so as not to obscure the different aspects of the embodiments. References to “an embodiment” in this description indicate that a particular configuration, structure or feature described in relation to the embodiment is included in at least one embodiment. Consequently, phrases such as “in one embodiment” that may appear at different points of the present description do not necessarily refer exactly to the same embodiment. Furthermore, specific formations, structures or features may be combined in any appropriate manner in one or more embodiments.

Embodiments of the present invention will be described in specific contexts, e.g., an adaptive TX-RX crosstalk cancellation for millimeter-wave radars operated as a frequency-modulated continuous wave (FMCW) radar. Embodiments of the present invention may be used with other types of radars systems, such as radars operating outside the millimeter-wave range.

In an embodiment of the present invention, a millimeter-wave radar adaptively cancels the effect of crosstalk from the transmit path to the receive path on a radar spectrum based on whether close-range targets are detected. A set of radar data is selected from a set of radar data. If no close-range targets are detected in the set of radar data, a crosstalk compensation factor is updated. If a close-range target is detected, the crosstalk compensation factor is not updated. The radar spectrum is compensated by cancelling the contribution of the crosstalk (e.g., in the time domain or in the frequency domain) based on the crosstalk compensation factor.

FIG. 1 shows a schematic diagram of millimeter-wave radar system 100, according to an embodiment of the present invention. Millimeter-wave radar system 100 includes millimeter-wave radar sensor 102 and processing system 104.

In some embodiments, millimeter-wave radar sensor 102 may be used to generate radar data ($x_{out_dig}(n)$) and processing system 104 may implement a radar processing chain, e.g., to detect, track, and/or classify targets.

During normal operation, millimeter-wave radar sensor 102 operates as a frequency-modulated continuous-wave (FMCW) radar sensor and transmits a plurality of radar signals 106, such as chirps, towards scene 120 using one or more transmitter (TX) antennas 114. The radar signals 106 are generated using RF and analog circuits 130. The radar signals 106 may be, e.g., in the 20 GHz to 122 GHz range. Other frequencies may also be used.

The objects in scene 120 may include one or more static or moving objects, such as cars, motorcycles, bicycles, trucks, and other vehicles, idle and moving humans and animals, poles, furniture, machinery, mechanical structures, walls and other types of structures. Other objects may also be present in scene 120.

The radar signals 106 are reflected by objects in scene 120. The reflected radar signals 108, which are also referred to as the echo signal, are received by one or more receiving (RX) antennas 116. RF and analog circuits 130 processes the received reflected radar signals 108 using, e.g., band-pass filters (BPFs), low-pass filters (LPFs), mixers, low-noise amplifier (LNA), and/or intermediate frequency (IF) amplifiers in ways known in the art to generate an analog signal $x_{out}(t)$.

The analog signal $x_{out}(t)$ is converted to raw digital data $x_{out_dig}(n)$ (also referred to as radar data) using ADC 112. The raw digital data $x_{out_dig}(n)$ is processed by processing system 104 to, e.g., detect, track, and/or classify targets.

Controller no controls one or more circuits of millimeter-wave radar sensor 102, such as RF and analog circuit 130 and/or ADC 112. Controller no may be implemented, e.g., as a custom digital or mixed signal circuit, for example. Controller no may also be implemented in other ways, such as using a custom or general purpose processor or controller configured to execute instructions stored in a memory. In some embodiments, processing system 104 implements a portion or all of controller no.

Processing system 104 may be implemented with a general purpose processor, controller or digital signal processor (DSP) that includes, for example, combinatorial circuits coupled to a memory. In some embodiments, processing system 104 may be implemented as an application specific integrated circuit (ASIC). In some embodiments, processing system 104 may be implemented with an ARM, RISC, or x86 architecture, for example. In some embodiments, processing system 104 may include an artificial intelligence (AI) accelerator. Some embodiments may use a combination of hardware accelerator and software running on a DSP or general purpose microcontroller. In some embodiments, processing system 104 may be implemented with a plurality of processors and/or controllers. Other implementations are also possible.

In some embodiments, millimeter-wave radar sensor 102 and a portion or all of processing system 104 may be implemented inside the same integrated circuit (IC). For example, in some embodiments, millimeter-wave radar sensor 102 and a portion or all of processing system 104 may be implemented in respective semiconductor substrates that are integrated in the same package. In other embodiments, millimeter-wave radar sensor 102 and a portion or all of processing system 104 may be implemented in the same monolithic semiconductor substrate. In some embodiments, millimeter-wave radar sensor 102 and processing system 104 are implemented in respective integrated circuits. In

some embodiments, a plurality of integrated circuits is used to implement millimeter-wave radar sensor **102**. In some embodiments, a plurality of integrated circuits is used to implement processing system **104**. Other implementations are also possible.

As a non-limiting example, RF and analog circuits **130** may be implemented, e.g., as shown in FIG. 1. During normal operation, voltage-controlled oscillator (VCO) **136** generates radar signals, such as a linear frequency chirps (e.g., from 57 GHz to 64 GHz, or from 76 GHz to 77 GHz), which are transmitted by the transmitting antenna **114**. The VCO **136** is controlled by PLL **134**, which receives a reference clock signal (e.g., 80 MHz) from reference oscillator **132**. PLL **134** is controlled by a loop that includes frequency divider **138** and amplifier **140**. Amplifiers **137** may be used to drive transmitting antenna **114**.

The TX radar signals **106** transmitted by transmitting antenna **114** are reflected by objects in scene **120** and received by receiving antenna **116**. The echo received by receiving antenna **116** is mixed with a replica of the signal transmitted by transmitting antenna **114** using mixer **146** to produce intermediate frequency (IF) signal $x_{IF}(t)$ (also known as beat signal). In some embodiments, the beat signal $x_{IF}(t)$ has a bandwidth between 10 kHz and 1 MHz. Beat signals with a bandwidth lower than 10 kHz or higher than 1 MHz is also possible. Amplifier **145** may be used to receive the reflected radar signals from antenna **116**.

Beat signal $x_{IF}(t)$ is filtered with low-pass filters (LPF) **148** and then sampled by ADC **112**. ADC **112** is advantageously capable of sampling the filtered beat signal $x_{out}(t)$ with a sampling frequency that is much smaller than the frequency of the signal received by receiving antenna **116**. Using FMCW radars, therefore, advantageously allows for a compact and low cost implementation of ADC **112**, in some embodiments.

The raw digital data $x_{out_dig}(n)$, which in some embodiments include the digitized version of the filtered beat signal $x_{out}(t)$ is (e.g., temporarily) stored, e.g., in matrices of $N_c \times N_s$ per receiving antenna **116**, where N_c is the number of chirps considered in a frame and N_s is the number of samples per chirp, for further processing by processing system **104**.

FIG. 2 shows a sequence of chirps **106** transmitted by TX antenna **114**, according to an embodiment of the present invention. As shown by FIG. 2, chirps **106** are organized in a plurality of frames and may be implemented as up-chirps. Some embodiments may use down-chirps or a combination of up-chirps and down-chirps, such as up-down chirps and down-up chirps. Other waveform shapes may also be used.

As shown in FIG. 2, each frame may include a plurality of chirps **106**. For example, in some embodiments, the number of chirps in a frame is 16. Some embodiments may include more than 16 chirps per frame, such as 20 chirps, 32 chirps, or more, or less than 16 chirps per frame, such as 10 chirps, 8 chirps, or less. In some embodiments, each frame includes only a single chirp.

In some embodiments, frames are repeated every FT time. In some embodiments, FT time is 50 ms. A different FT time may also be used, such as more than 50 ms, such as 60 ms, 100 ms, 200 ms, or more, or less than 50 ms, such as 45 ms, 40 ms, or less.

In some embodiments, the FT time is selected such that the time between the beginning of the last chirp of frame n and the beginning of the first chirp of frame $n+1$ is equal to PRT. Other embodiments may use or result in a different timing.

The time between chirps of a frame is generally referred to as pulse repetition time (PRT). In some embodiments, the

PRT is 5 ms. A different PRT may also be used, such as less than 5 ms, such as 4 ms, 2 ms, or less, or more than 5 ms, such as 6 ms, or more.

The duration of the chirp (from start to finish) is generally referred to as chirp time (CT). In some embodiments, the chirp time may be, e.g., 64 μ s. Higher chirp times, such as 128 μ s, or higher, may also be used. Lower chirp times, may also be used.

In some embodiments, the chirp bandwidth may be, e.g., 4 GHz. Higher bandwidth, such as 6 GHz or higher, or lower bandwidth, such as 2 GHz, 1 GHz, or lower, may also be possible.

In some embodiments, the sampling frequency of millimeter-wave radar sensor **902** may be, e.g., 1 MHz. Higher sampling frequencies, such as 2 MHz or higher, or lower sampling frequencies, such as 500 kHz or lower, may also be possible.

In some embodiments, the number of samples used to generate a chirp may be, e.g., 64 samples. A higher number of samples, such as 128 samples, or higher, or a lower number of samples, such as 32 samples or lower, may also be used.

As signal transmission and reception is performed simultaneously, crosstalk may occur between transmit path **113** and receive path **115**. Crosstalk from transmit path **113** to receive path **115** generally covers the lower part of the range spectrum with high amplitudes. For example, FIG. 3 shows exemplary range spectrum **300** associated with millimeter-wave radar sensor **102**, according to an embodiment of the present invention. Range spectrum **300** is obtained from a fast Fourier transform (FFT) and non-coherent integration of all chirps and all channels of the radar data ($x_{out_dig}(n)$) of a single frame with $N_s=128$, $N_c=16$, $N_R=8$ and $BW=4$ GHz, where N_c is the number of chirps considered in a frame, N_s is the number of samples per chirp, and N_R is the number of receiving antennas **116**, where scene **120** is empty (without any targets present) and with transmitting antenna(s) **114** and receiving antenna(s) **116** located at 1 meter from the floor.

As shown in FIG. 3, the first range bins (bin **27** and lower) of the range spectrum **300** are dominated by the TX-RX crosstalk. The low frequency components of the TX-RX crosstalk are caused by the direct transmission from the transmit path **113** to the receive path **115**, which leads to a very small round-trip delay and thus a very small beat frequency of the beat signal $x_{IF}(t)$. The reflections that are present after range bin **27** are due to reflections from the ground, e.g. at approximately 1 meter distance due to the 1 meter height of the radar above the ground.

As illustrated in FIG. 3, the TX-RX crosstalk may be a dominant low-frequency component in the receive signal, which may be modelled as target with a spread function $g_0(T)$ with a bounded support on $[0, \tau_L]$ with $\tau_L \ll 1$ as the crosstalk covers the lower part of the spectrum only, where L correspond to the bin representing the highest bin associated with the TX-RX crosstalk (bin **27** in the example illustrated in FIG. 3), τ denoting the normalized beat frequency, where $\tau \in (0, 1]$, and where $\tau=1$ corresponds to the maximum unambiguous range.

As illustrated in FIG. 3, the TX-RX crosstalk may cause the detection of static objects at close ranges to be extremely challenging. This is particularly the case if the targets have a comparably small radar cross-section (RCS). For example, FIG. 4 shows exemplary range spectrum **400** associated with millimeter-wave radar sensor **102**, according to an embodiment of the present invention. Range spectrum **400** is obtained in a similar manner as range spectrum **300**, except

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that scene **120** includes a plastic pole instead of an empty scene. As shown in FIG. 4, detecting the plastic pole target, which is located at a distance that corresponds to bin **17**, may be difficult in the presence of the TX-RX crosstalk.

FIG. 5 shows waveforms **502** and **504** of radar data $x_{out_dig}(n)$ associated with an empty scene, and with a scene with a target at 1 meter, respectively, according to an embodiment of the present invention.

As illustrated by curve **502** (and as also illustrated in FIG. 3), the contribution of the TX-RX crosstalk has a low frequency. As illustrated by curve **504** (and as also illustrated in FIG. 4), when a target at close-range is present (e.g., in the field-of-view of millimeter-wave radar sensor **102** and close to the millimeter-wave radar sensor, such as within 1 meter), the reflected signals (**108**) are superimposed with the TX-RX crosstalk.

As illustrated in FIGS. 3-5, it may be desirable to compensate for TX-RX crosstalk, e.g., before performing other radar processing tasks, such as target detection, tracking, and/or classification. FIG. 6 shows a flow chart of embodiment method **600** for dynamic TX-RX crosstalk cancellation, according to an embodiment of the present invention. Method **600** may be implemented by processing system **104**.

During step **602**, a millimeter-wave radar sensor (e.g., **102**), transmits radar signals (e.g., **106**) organized in frames using one or more transmitting antennas (e.g., **114**).

During step **604**, the millimeter-wave radar sensor receives signals using one or more receiving antennas (e.g., **116**), which may include reflected radar signals (e.g., **108**). The received signals may be organized in frames corresponding to the frames transmitted during step **602**.

During step **606**, raw digital data $x_{out_dig}(n)$ is generated based on the received signals (which includes TX-RX crosstalk and may include reflected radar signals **108**). For example, in some embodiments, during step **606**, the transmitted and received signals are mixed (e.g., using mixer **146**) to generate an IF signal $x_{IF}(t)$. The IF signal $x_{IF}(t)$ is then filtered (e.g., with a low-pass and/or band-pass filter) and digitized with an ADC (e.g., **112**) to generate the radar data $x_{out_dig}(n)$.

During step **610**, a set X_S of the radar data $x_{out_dig}(n)$ is selected. For example, in some embodiments, the set X_S of radar data $x_{out_dig}(n)$ corresponds to a selection $\mathcal{S}$ of chirps with $|\mathcal{S}|=S \in \{1, \dots, N_C N_R\}$ from radar data $x_{out_dig}(n)$, where $X_S \in \mathbb{R}^{N_S \times S}$ denote the chirps corresponding to the selection $\mathcal{S}$ arranged in a matrix. In some embodiments, the set X_S includes all radar data $x_{out_dig}(n)$ (the selection $\mathcal{S}$ of chirps includes all chirps in the radar data $x_{out_dig}(n)$). In some embodiments, the set X_S includes some, but not all, radar data $x_{out_dig}(n)$.

During step **612**, a crosstalk approximation of the TX-RX crosstalk is determined, e.g., by using a least-squares fit of a low-order polynomial performed on the set X_S of radar data $x_{out_dig}(n)$. As illustrated by curve **502**, in some embodiments, such least-squares fit may represent a good approximation of the TX-RX crosstalk. In some embodiments, a polynomial of order lower than 10, such as 6, may be used.

For example, in some embodiments, the least-squares fit is performed for each column of X_S . For example, without loss of generality, letting the sampling times be $n=1, \dots, N_S$, and considering a polynomial of degree d given as

$$p_d(n;a) = \sum_{k=0}^d a(k)n^k, \quad (1)$$

where $a=[a(0), \dots, a(d)] \in \mathbb{R}^d$ with $d=\bar{d}+1$ and $a \in \mathbb{R}^d$ parameterizes the polynomial, where

$$p_d(a)=[p_d(1;a), \dots, p_d(N_S;a)] \in \mathbb{R}^{N_S} \quad (2)$$

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and

$$P_d(A)=[p_d(a_1), \dots, p_d(a_S)] \in \mathbb{R}^{N_S \times S} \quad (3)$$

with

$$A=[a_1, \dots, a_S] \in \mathbb{R}^{d \times S} \quad (4)$$

the least-square problem may be given as

$$\min_A \|X_S - P_d(A)\|_F, \quad (5)$$

where $\|\cdot\|_F$ represents the Frobenius norm. Letting the matrix $B \in \mathbb{R}^{N_S \times d}$ be a Vandermonde matrix with elements $B(i,j) = i^{j-1}$ such that $p_d(a) = Ba$ and $P_d(A) = BA$, the linear least-squares problem may be given as

$$\min_A \|X_S - BA\|_F, \quad (6)$$

which as $B \in \mathbb{R}^{N_S \times d}$ has full rank has a unique optimizer if $d \leq N_S$ given as

$$A_{opt} = (B'B)^{-1}B'X_S \quad (7)$$

and a unique optimal value, where B' refers to the transpose of B if B is a real matrix and to the conjugate transpose (Hermitian) of B if B is a complex matrix. The fitted polynomials may be given as

$$P_d(A_{opt}) = B(B'B)^{-1}B'X_S = P_B X_S \quad (8)$$

and may be obtained by the orthogonal projection $P_B X_S$ of X_S onto $\text{range}(B)$.

During step **614**, the computation of a residual of the estimation of the TX-RX crosstalk is performed (the computation of the residual of the crosstalk approximation performed during step **612**). For example, in some embodiments, the residual of the polynomial fit (computed during step **612**) is computed, e.g., by subtracting the polynomial fit from the set X_S . For example, in some embodiments, the residual X_R may be computed as

$$X_R = X_S - P_d(A_{opt}) = P_B^\perp X_S, \quad (9)$$

which corresponds to the projection of X_S onto $(\text{range}(B))^\perp$, where $B'B$ as well as $B'X_S$ can be obtained in closed form.

As illustrated in FIG. 5, the residual X_R may be indicative of the presence of close-range target. During step **616**, a spectrum is determined based on residual X_R using a spectral estimator. For example, in some embodiments, considering real-valued raw ADC data X_S (although the approach may not be limited to real-valued raw ADC data and may be applied, e.g., to real and imaginary part of complex-valued ADC data), the range spectrum of X_S is symmetric by construction and only one half of the range spectrum contains useful information. Thus, the spectral estimator may be defined as a function mapping from $\mathbb{R}^{N_S \times S}$ to

$$\mathbb{C}^{\frac{q}{2} N_S \times S},$$

only, where the additional factor q accounts for spectral estimators which estimate the spectrum on, e.g., an over-sampled FFT grid. For example, in some embodiments, the spectral estimator $\mathcal{F}_R$ (where

$$\mathcal{F}_R: \mathbb{R}^{N_S \times S} \rightarrow \mathbb{C}^{\frac{q}{2}N_S \times S}, X_R \mapsto \mathcal{F}_R(X_R)$$

estimates the useful part of the range spectrum of X_R using a spectral grid with

$$\frac{q}{2}N_S$$

grid points and may generate the range spectrum using an FFT (with time-domain windowing) along the first dimension of X_R and by considering the first half of the FFT spectrum only. In some embodiments, the spectra P_R of the residual X_R may be given by

$$P_R = \mathcal{F}_R(X_R) \quad (10)$$

During step **618**, the first L bins of the spectra P_R of the residual X_R are integrated. For example, in some embodiments, the first L bins of the range spectra P_R of the residuum along all S is incoherently (non-coherently) integrated to obtain the spectrum $P_D \in \mathbb{R}^L$, which may be computed as

$$P_D = (P_R(1:L,:)) \odot P_R^*(1:L,:) \mathbf{1}_{S \times 1} \in \mathbb{R}^L, \quad (11)$$

where $\odot$ denotes the element-wise Hadamard matrix product, P_R^* denotes the complex conjugate of P_R , and $\mathbf{1}_{S \times 1}$ denotes the all-ones vector of dimension S , and where

$$L(0 < L \ll \frac{q}{2}N_S)$$

corresponds to the last range bin that is significantly affected by the TX-RX crosstalk.

In some embodiments, spectrum P_D is indicative of whether a close-range target is present. During step **620**, target detection is performed, e.g., using constant false alarm rate (CFAR). In some embodiments, the CFAR may be aborted after the first target is detected.

If no target is detected during step **622**, a crosstalk compensation factor P_{est} is updated during step **624** (where the crosstalk compensation factor P_{est} may be previously initialized, e.g., to a default value). For example, let $P_{est} \in \mathbb{C}^{L \times N_C \times N_R}$ denote the current estimate of the TX-RX crosstalk spectrum (e.g., which is stored in a memory of processing system **104**). In addition, let

$$\mathcal{F}: \mathbb{R}^{N_S \times N_C \times N_R} \rightarrow \mathbb{C}^{\frac{q}{2}N_S \times N_C \times N_R}, X \mapsto \mathcal{F}(X)$$

denote a spectral estimator which estimates the range spectra of every chirp and receive channel in the radar data X of the current frame, the estimate of the spectrum of the TX-RX crosstalk may be updated according to an exponential moving average (EMA) given as

$$P_{est} \leftarrow \alpha P_{est} (1 - \alpha) P(1:L, :, :), \quad (12)$$

where $P(1:L, :, :)$ denotes the first L range bins of the estimated range spectra of the raw ADC data $x_{out_dig}(n)$ (of $P = \mathcal{F}(X)$, where X represents the radar data $x_{out_dig}(n)$) and α with $0 \leq \alpha \leq 1$ denotes the exponential weighting factor of the EMA, and where $\leftarrow$ is a symbol that implies that the operation to the right of the symbol $\leftarrow$ is performed first and then assigned to the variable to the left of the symbol $\leftarrow$. The updated crosstalk compensation factor P_{est} may be stored

(e.g., replacing the previous crosstalk compensation factor P_{est}) in (e.g., volatile or non-volatile) memory (e.g., of processing system **104**).

If a target is detected during step **622**, step **624** is skipped.

During step **626** (which may be performed before, during, or after steps **608** and **624** (if applicable)), the spectrum of the radar data is estimated. For example, in some embodiments, the spectrum P of the radar data $x_{out_dig}(n)$ may be given as

$$P = \mathcal{F}(X) \quad (13)$$

where $\mathcal{F}(\cdot)$ is the same spectral estimator used during step **616** and where X represents the radar data $x_{out_dig}(n)$. In some embodiments, the spectral estimator $\mathcal{F}(\cdot)$ is a range FFT. In some embodiments, the spectral estimator $\mathcal{F}(\cdot)$ is a Capon estimator. Other estimators may also be used for performing steps **626** and **616**.

During step **628**, the spectrum generated during step **626** is compensated based on the stored crosstalk compensation factor P_{est} . For example, in some embodiments, the spectrum P generated during step **626** is compensated as

$$P(1:L, :, :) \leftarrow P(1:L, :, :) - P_{est} \quad (14)$$

In some embodiments, L may be, e.g., $qN_S/2$. In some embodiments, the compensated spectrum generated during step **626** is advantageously not affected (or not substantially affected) by TX-RX crosstalk.

During step **630**, the compensated spectrum P is used, e.g., to detect, track and/or classify targets in ways known in the art.

As shown in FIG. **6**, step **608** (which includes **610**, **612**, **614**, **616**, **618**, **620**, and **622**) operates as a classifier that classifies X_S as including TX-RX crosstalk or including TX-RX crosstalk plus target(s). As also illustrated in FIG. **6**, the crosstalk compensation factor P_{est} (which is indicative of the TX-RX crosstalk between transmit path **113** and receive path **115**) is updated (in step **624**) only when no target is detected in the chirps of set X_S . By only updating the crosstalk compensation factor P_{est} when no target is detected, some embodiments advantageously compensate for TX-RX crosstalk without including contributions from close-range target(s), increasing the performance of the crosstalk cancellation.

As shown in FIG. **6**, in some embodiments, step **618** may be performed to generate an integrated spectrum of the residual by performing incoherent integration. In some embodiments, coherent integration may be performed instead (e.g., with an additional beamforming step). In some embodiments, step **618** may be omitted and step **620** may be performed in a non-integrated spectrum of the residual (e.g., generated during step **618**).

In some embodiments, step **608** and step **624** (if applicable) are performed with the same rate as the frame rate of the radar data. By performing TX-RX crosstalk cancellation at the same rate as the frame rate, some embodiments advantageously adapt to new environments quickly. For example, in some embodiments, millimeter-wave radar sensor **102** may be implemented in a mobile device (such as a smartphone) and may move quickly, such as by rotating or displacing. A fast TX-RX crosstalk cancellation rate may advantageously allow for close-range target detection, which may improve the performance of, e.g., hand gesture recognition for operating the smartphone.

In some embodiments, step **608** and step **624** (if applicable) are performed with rate slower than the frame rate of the radar data (such as 10 times slower or 100 times slower). By performing TX-RX crosstalk cancellation at a rate

slower than the frame rate, some embodiments advantageously save computational power. For example, in some embodiments, the environment influencing the TX-RX crosstalk may change at a lower rate. For example, a millimeter-wave radar sensor that is not mobile and is placed in an indoor environment (such as at the ceiling of a room) may not experience fast changes in the TX-RX crosstalk (for example, changes due to temperature may have a slow time constant). Thus, some embodiments may advantageously save computational power by using a slow rate of TX-RX crosstalk estimation without substantially impacting the TX-RX crosstalk cancellation performance. In some embodiments, the TX-RX crosstalk cancellation (e.g., step 628) may be performed at a faster rate (e.g., every frame) than the rate of performing the TX-RX crosstalk estimation (e.g., step 608 and 624 (if applicable)).

In some embodiments, the TX-RX crosstalk cancellation rate may be dynamically changed. For example, in an embodiment in which millimeter-wave radar sensor 102 is implemented in a vehicle, the TX-RX crosstalk cancellation rate may be based on the speed of the vehicle. For example, when the vehicle is stopped at a red light, the TX-RX crosstalk cancellation rate may be slow (e.g., 10 times or 100 times slower than the radar frame rate). Having a slow TX-RX crosstalk cancellation rate when the vehicle is stopped may advantageously save computational power and battery power without substantially impacting the TX-RX crosstalk cancellation performance (since the environment may not change very rapidly). When the vehicle is moving at a high speed in a highway, the TX-RX crosstalk cancellation rate may be fast (e.g., at the same rate as the frame rate of the radar), which may advantageously improve the performance of the TX-RX crosstalk cancellation as the vehicle moves and turns at high speed).

Advantages of some applications include improved target detection performance for close-range targets, e.g., when compared with radars that use HPFs to address crosstalk (which may introduce a minimum range constraint). Additional advantages include the ability to cancel TX-RX crosstalk in applications for detecting static targets.

As shown in FIG. 6, the TX-RX crosstalk compensation may be performed in the frequency domain (as illustrated by step 628). In some embodiments, the TX-RX crosstalk compensation may be performed in the time domain. For example, FIG. 7, shows a flow chart of embodiment method 700 for dynamic TX-RX crosstalk cancellation, according to an embodiment of the present invention. Method 700 may be implemented by processing system 104. Method 700 is similar and may be performed in a similar manner as method 600. Method 700, however, includes steps 724 and 728 for compensating for TX-RX crosstalk in the time domain, and step 626 is performed on the compensated time-domain radar data $x_{out_dig_comp}(n)$ instead of on $x_{out_dig}(n)$.

For example, during step 724 (when no target is detected in the set S of the radar data $x_{out_dig}(n)$), a crosstalk compensation factor $X_{crosstalk}$ is updated, where $X_{crosstalk}$ is a time domain compensation factor ($X_{crosstalk} \in \mathbb{R}^{N_S \times N_C \times N_R}$) and where the compensation factor $X_{crosstalk}$ may be previously initialized, e.g., to a default value. For example, in some embodiments, $X_{crosstalk}$ corresponds to the best fit curve generated during step 612, and which may substantially approximate the TX-RX crosstalk. Updating the crosstalk compensation factor $X_{crosstalk}$ may be performed using EMA, e.g., as

$$X_{crosstalk} \leftarrow \alpha X_{crosstalk} + (1 - \alpha) \mathcal{D}(P_d(A_{opt})), \quad (15)$$

where $\mathcal{D} : \mathbb{R}^{N_S \times S} \rightarrow \mathbb{R}^{N_S \times N_C \times N_R}$, $P_d(A_{opt}) \mapsto \mathcal{D}(P_d(A_{opt}))$ maps the least-squares fit $P_d(A_{opt})$ based on the selection S to the corresponding elements in $X_{crosstalk}$. In some embodiments, the selection S contains at least one chirp of each receive channel to obtain a reasonable mapping $\mathcal{D}$.

In some embodiments, the compensated radar data (comp) may be given as

$$X_{comp} = X - X_{crosstalk} \quad (16)$$

where X represents the radar data $x_{out_dig}(n)$.

During step 730, the spectrum of the compensated radar data (comp) is performed, e.g., in a similar manner as in step 730. For example, in some embodiments, the spectrum of the compensated radar data is performed as

$$P = \mathcal{F}(X_{comp}) \quad (17)$$

where $\mathcal{F}(\cdot)$ is the same spectral estimator used during step 616.

Advantages of some embodiment implementing the TX-RX cancellation in the time domain (e.g., method 700) include achieving TX-RX cancellation without suffering from residual side lobes of the TX-RX crosstalk.

Example embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.

Example 1. A method including: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; and after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

Example 2. The method of example 1, further including: determining a crosstalk approximation of the transmitter-receiver crosstalk on the set of data; determining a residual of the crosstalk approximation based on the polynomial fit and the set of data; and determining a spectrum of the residual of the crosstalk approximation, where performing the target detection on the set of data includes performing the target detection on the set of data based on the spectrum of the residual of the polynomial fit.

Example 3. The method of examples 1 or 2, further including: determining a polynomial fit on the set of data; determining a residual of the polynomial fit based on the polynomial fit and the set of data; and determining a spectrum of the residual of the polynomial fit, where performing the target detection on the set of data includes performing the target detection on the set of data based on the spectrum of the residual of the polynomial fit.

Example 4. The method of one of examples 1 to 3, where determining the polynomial fit includes performing a least-squares fit of low-order polynomial on the set of data, where the low-order polynomial is a polynomial of order between 6 and 10.

Example 5. The method of one of examples 1 to 4, further including integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower

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than or equal to a maximum number of bins of the spectrum of the residual of the polynomial fit, where performing the target detection on the set of data includes performing the target detection on the set of data based on the integrated spectrum.

Example 6. The method of one of examples 1 to 5, where integrating the first L bins of the spectrum of the residual of the polynomial fit includes incoherently integrating the first L bins of the spectrum of the residual of the polynomial fit.

Example 7. The method of one of examples 1 to 6, where generating the radar spectrum and generating the spectrum of the residual of the polynomial fit includes using the same spectral estimator.

Example 8. The method of one of examples 1 to 7, where selecting the set of data and performing the target detection on the set of data includes selecting the set of data and performing the target detection on the set of data at a same rate as a frame rate of the radar data.

Example 9. The method of one of examples 1 to 7, where selecting the set of data and performing the target detection on the set of data includes selecting the set of data and performing the target detection on the set of data at a rate lower than a frame rate of the radar data.

Example 10. The method of one of examples 1 to 7, where selecting the set of data and performing the target detection on the set of data includes selecting the set of data and performing the target detection on the set of data at a dynamic rate.

Example 11. The method of one of examples 1 to 10, further including generating the radar spectrum based on the radar data and the crosstalk compensation factor at a faster rate than performing the target detection on the set of data.

Example 12. The method of one of examples 1 to 11, where performing the target detection on the set of data includes performing the target detection on the set of data using a constant false alarm rate (CFAR).

Example 13. The method of one of examples 1 to 12, further including aborting the target detection on the set of data when a first target is detected.

Example 14. The method of one of examples 1 to 13, where updating the crosstalk compensation factor includes updating the crosstalk compensation factor using an exponential moving average.

Example 15. The method of one of examples 1 to 14, where the crosstalk compensation factor includes a crosstalk spectrum, and where generating the radar spectrum includes: generating a first radar spectrum based on the radar data; and subtracting the crosstalk spectrum from the first radar spectrum.

Example 16. The method of one of examples 1 to 14, where the crosstalk compensation factor includes a time-domain crosstalk estimate, and where generating the radar spectrum includes: subtracting the time-domain crosstalk estimate from the radar data to generate a compensated radar data; and generating the radar spectrum based on the compensated radar data.

Example 17. The method of one of examples 1 to 16, where the set of data includes a plurality of linear chirps.

Example 18. The method of one of examples 1 to 17, further including generating the radar data based on signals from the receive path of the radar sensor.

Example 19. The method of one of examples 1 to 18, further including performing target detection or target tracking based on the radar spectrum.

Example 20. The method of one of examples 1 to 19, where the radar sensor is a millimeter-wave radar sensor.

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Example 21. The method of one of examples 1 to 20, where the set of data is a sub-set of radar data selected from the radar data.

Example 22. A radar system including: a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of the receive path; and a processing system configured to: initialize a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path, select a set of data from the radar data, perform target detection on the set of data, and after performing the target detection on the set of data, when no target is detected in the set of data, update the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generate a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generate the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

Example 23. A method including: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the a transmit path of a millimeter-wave radar sensor and a receive path of the millimeter-wave radar sensor; receiving radar data from the millimeter-wave radar sensor; selecting a sub-set of data from the radar data; determining a polynomial fit on the sub-set of data; determining a residual of the polynomial fit based on the polynomial fit and the sub-set of data; and determining a spectrum of the residual of the polynomial fit; incoherently integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than a maximum number of bins of the spectrum of the residual of the polynomial fit; performing target detection on the sub-set of data based on the integrated spectrum; after performing the target detection on the sub-set of data, when no target is detected in the sub-set of data, updating the crosstalk compensation factor based on the sub-set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the sub-set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; and performing target detection or target tracking based on the radar spectrum.

Example 24. A method including: receiving radar data from a radar sensor; selecting a set of data from the radar data; determining a crosstalk approximation of a transmitter-receiver crosstalk between a transmit path of the radar sensor and a receive path of the radar sensor based on the set of data; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; determining a spectrum of the residual of the crosstalk approximation; and performing target detection on the set of data based on the spectrum of the residual of the crosstalk approximation.

Example 25. The method of example 24, where determining the crosstalk approximation includes determining a polynomial fit on the set of data, and where determining the residual of the crosstalk approximation includes subtracting the crosstalk approximation from the set of data.

While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as

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other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

What is claimed is:

1. A method comprising:

initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor;

receiving radar data from the radar sensor;

selecting a set of data from the radar data;

performing target detection on the set of data;

after performing the target detection on the set of data,

when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and

when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor;

determining a crosstalk approximation of the transmitter-receiver crosstalk;

determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; and

determining a spectrum of the residual of the crosstalk approximation, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the spectrum of the residual of the crosstalk approximation, and the spectrum of the residual of the crosstalk approximation comprises range bins representing frequency domain values.

2. The method of claim 1, wherein determining the crosstalk approximation comprises determining a polynomial fit on the set of data, wherein determining the residual of the crosstalk approximation comprises determining a residual of the polynomial fit based on the polynomial fit and the set of data, and wherein determining the spectrum of the residual of the crosstalk approximation comprises determining a spectrum of the residual of the polynomial fit.

3. The method of claim 2, wherein determining the polynomial fit comprises performing a least-squares fit of a low-order polynomial on the set of data, wherein the low-order polynomial is a polynomial of order between 6 and 10.

4. The method of claim 2, further comprising integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than or equal to a maximum number of bins of the spectrum of the residual of the polynomial fit, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the integrated spectrum.

5. The method of claim 4, wherein integrating the first L bins of the spectrum of the residual of the polynomial fit comprises incoherently integrating the first L bins of the spectrum of the residual of the polynomial fit.

6. The method of claim 1, wherein:

generating the radar spectrum comprises using a spectral estimator; and

generating the spectrum of the residual of the crosstalk approximation comprises using the spectral estimator.

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7. The method of claim 1, wherein performing the target detection on the set of data comprises performing the target detection on the set of data using a constant false alarm rate (CFAR).

8. The method of claim 1, further comprising aborting the target detection on the set of data when a first target is detected.

9. The method of claim 1, wherein updating the crosstalk compensation factor comprises updating the crosstalk compensation factor using an exponential moving average.

10. The method of claim 1, wherein the crosstalk compensation factor comprises a crosstalk spectrum, and wherein generating the radar spectrum comprises:

generating a first radar spectrum based on the radar data; and

subtracting the crosstalk spectrum from the first radar spectrum.

11. The method of claim 1, wherein the crosstalk compensation factor comprises a time-domain crosstalk estimate, and wherein generating the radar spectrum comprises:

subtracting the time-domain crosstalk estimate from the radar data to generate compensated radar data; and

generating the radar spectrum based on the compensated radar data.

12. The method of claim 1, wherein the set of data comprises a plurality of linear chirps.

13. The method of claim 1, further comprising generating the radar data based on signals from the receive path of the radar sensor.

14. The method of claim 1, further comprising performing the target detection or target tracking based on the radar spectrum.

15. The method of claim 1, wherein the radar sensor is a millimeter-wave radar sensor.

16. The method of claim 1, wherein the set of data is a subset of radar data selected from the radar data.

17. The method of claim 1, wherein:

selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a same rate as a frame rate of the radar data; or

selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a rate lower than the frame rate of the radar data; or

selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a dynamic rate; or

the method further comprises generating the radar spectrum based on the radar data and the crosstalk compensation factor at a faster rate than performing the target detection on the set of data.

18. A radar system comprising:

a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of the receive path; and

at least one processor and at least one memory with instructions stored thereon, wherein the instructions, when executed by the at least one processor enable the radar system to perform the steps of:

initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path,

selecting a set of data from the radar data,

performing target detection on the set of data,

after performing the target detection on the set of data,

when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and

when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor;

determining a crosstalk approximation of the transmitter-receiver crosstalk;

determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; and

determining a spectrum of the residual of the crosstalk approximation, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the spectrum of the residual of the crosstalk approximation, and the spectrum of the residual of the crosstalk approximation comprises range bins representing frequency domain values.

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performing target detection on the set of data, and
 after performing the target detection on the set of data,
 when no target is detected in the set of data, updating
 the crosstalk compensation factor based on the set
 of data and, after updating the crosstalk compen- 5
 sation factor, generating a radar spectrum based on
 the radar data and the crosstalk compensation
 factor, and
 when a target is detected in the set of data, generating
 the radar spectrum based on the radar data and the 10
 crosstalk compensation factor without updating
 the crosstalk compensation factor;
 determining a crosstalk approximation of the trans-
 mitter-receiver crosstalk; determining a residual
 of the crosstalk approximation based on the cross-
 talk approximation and the set of data; and
 determining a spectrum of the residual of the cross-
 talk approximation, wherein performing the target
 detection on the set of data comprises performing
 the target detection on the set of data based on the
 spectrum of the residual of the crosstalk approxi-
 mation, and the spectrum of the residual of the
 crosstalk approximation comprises range bins rep-
 resenting frequency domain values.

19. The radar system of claim 18, wherein:
 selecting the set of data and performing the target detec-
 tion on the set of data comprises selecting the set of
 data and performing the target detection on the set of
 data at a same rate as a frame rate of the radar data, or
 selecting the set of data and performing the target detec-
 tion on the set of data comprises selecting the set of
 data and performing the target detection on the set of
 data at a rate lower than the frame rate of the radar data,
 or
 selecting the set of data and performing the target detec-
 tion on the set of data comprises selecting the set of
 data and performing the target detection on the set of
 data at a dynamic rate, or
 the instructions, when executed by the at least one pro- 40
 cessor further enable the radar system to perform the
 step of generating the radar spectrum based on the radar
 data and the crosstalk compensation factor at a faster
 rate than performing the target detection on the set of
 data.

20. A method comprising:
 initializing a crosstalk compensation factor indicative of a
 transmitter-receiver crosstalk between a transmit path
 of a radar sensor and a receive path of the radar sensor;
 receiving radar data from the radar sensor; 50
 selecting a set of data from the radar data;
 performing target detection on the set of data;
 after performing the target detection on the set of data,
 when no target is detected in the set of data, updating
 the crosstalk compensation factor based on the set of 55
 data and, after updating the crosstalk compensation
 factor, generating a radar spectrum based on the
 radar data and the crosstalk compensation factor, and
 when a target is detected in the set of data, generating
 the radar spectrum based on the radar data and the 60
 crosstalk compensation factor without updating the
 crosstalk compensation factor;
 determining a crosstalk approximation of the transmitter-
 receiver crosstalk;
 determining a residual of the crosstalk approximation 65
 based on the crosstalk approximation and the set of
 data; and

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determining a spectrum of the residual of the crosstalk
 approximation, wherein performing the target detection
 on the set of data comprises performing the target
 detection on the set of data based on the spectrum of the
 residual of the crosstalk approximation, wherein deter-
 mining the crosstalk approximation comprises deter-
 mining a polynomial fit on the set of data, wherein
 determining the residual of the crosstalk approximation
 comprises determining a residual of the polynomial fit
 based on the polynomial fit and the set of data, and
 wherein determining the spectrum of the residual of the
 crosstalk approximation comprises determining a spec-
 trum of the residual of the polynomial fit.

21. The method of claim 20, wherein determining the
 polynomial fit comprises performing a least-squares fit of a
 low-order polynomial on the set of data, wherein the low-
 order polynomial is a polynomial of order between 6 and 10.

22. The method of claim 20, further comprising integrat-
 ing a first L bins of the spectrum of the residual of the
 polynomial fit to generate an integrated spectrum, L being a
 positive integer greater than 1 and lower than or equal to a
 maximum number of bins of the spectrum of the residual of
 the polynomial fit, wherein performing the target detection
 on the set of data comprises performing the target detection
 on the set of data based on the integrated spectrum.

23. The method of claim 22, wherein integrating the first
 L bins of the spectrum of the residual of the polynomial fit
 comprises incoherently integrating the first L bins of the
 spectrum of the residual of the polynomial fit.

24. A radar system comprising:

a radar sensor configured to transmit radar signals via a
 transmit path, receive reflected radar signals via a
 receive path, and provide radar data based on an output
 of the receive path; and

at least one processor and at least one memory with
 instructions stored thereon, wherein the instructions,
 when executed by the at least one processor enable the
 radar system to perform the steps of:

initializing a crosstalk compensation factor indicative
 of a transmitter-receiver crosstalk between the trans-
 mit path and the receive path,

selecting a set of data from the radar data,

performing target detection on the set of data,

after performing the target detection on the set of data,
 when no target is detected in the set of data, updating
 the crosstalk compensation factor based on the set of
 data and, after updating the crosstalk compensation
 factor, generating a radar spectrum based on the
 radar data and the crosstalk compensation factor, and
 when a target is detected in the set of data, generating
 the radar spectrum based on the radar data and the
 crosstalk compensation factor without updating the
 crosstalk compensation factor,

determining a crosstalk approximation of the transmit-
 ter-receiver crosstalk;

determining a residual of the crosstalk approximation
 based on the crosstalk approximation and the set of
 data, and

determining a spectrum of the residual of the crosstalk
 approximation, wherein performing the target detec-
 tion on the set of data comprises performing the
 target detection on the set of data based on the
 spectrum of the residual of the crosstalk approxima-
 tion, wherein determining the crosstalk approxima-
 tion comprises determining a polynomial fit on the
 set of data, wherein determining the residual of the
 crosstalk approximation comprises determining a

residual of the polynomial fit based on the polynomial fit and the set of data, and wherein determining the spectrum of the residual of the crosstalk approximation comprises determining a spectrum of the residual of the polynomial fit.

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25. The radar system of claim **24**, wherein determining the polynomial fit comprises performing a least-squares fit of a low-order polynomial on the set of data, wherein the low-order polynomial is a polynomial of order between 6 and 10.

26. The radar system of claim **24**, wherein the instructions, when executed by the at least one processor further enable the radar system to perform the step of integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than or equal to a maximum number of bins of the spectrum of the residual of the polynomial fit, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the integrated spectrum.

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